# SPA617

## Experiment 1 : RF Transmission, Reflections, and Standing Waves

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# Experiment 1 : RF Transmission, Reflections, and Standing Waves

## 1.1 Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

## 1.2 Objectives

(a) To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.

(b) To study signal reflection from an open-ended cable and from a cable terminated with a $50\ \Omega$ load.

(c) To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a $50\ \Omega$ termination, indicating that its characteristic impedance is close to $50\ \Omega$.

(d) To calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 \approx 50\ \Omega$.

(e) To measure the insertion loss and attenuation per unit length of the coaxial cable.

## 1.3 Theory

### 1.3.1 Transmission lines and reflections

A transmission line is a structure that guides electromagnetic energy from one point to another. A coaxial cable supports transverse electromagnetic (TEM) propagation, for which both the electric and magnetic fields are transverse to the direction of propagation. The signal velocity in a cable is lower than the speed of light in vacuum because of the dielectric material between the conductors. When a travelling wave encounters a load impedance $Z_L$ that is different from the characteristic impedance $Z_0$ of the cable, a part of the wave is reflected toward the source. At any fixed observation point, the incident and reflected waves superimpose. Depending on their relative phase, the measured amplitude may be large because of constructive interference or small because of destructive interference. In this experiment, the oscilloscope measures the voltage at one fixed point, namely the Tee junction. It does not directly display the spatial standing-wave pattern along the cable. When the signal frequency is

varied, the round-trip phase of the reflected wave changes, causing the measured amplitude at the Tee junction to vary periodically.

### 1.3.2 Phase velocity measurement

Let the physical length of the coaxial cable be $\ell$ and the phase velocity in the cable be $v_p$. The reflected signal travels a round-trip distance $2\ell$, so the round-trip time delay is

$$\tau = \frac{2\ell}{v_p}.$$

The round-trip phase of the reflected signal is

$$\phi_{\mathrm{RT}} = 2\pi f\tau + \phi_\Gamma = \frac{4\pi f\ell}{v_p} + \phi_\Gamma,$$

where $\phi_\Gamma$ is the phase shift introduced by the termination. For an ideal open circuit, $\phi_\Gamma = 0$.

Two consecutive maxima, or two consecutive minima, are separated by a round-trip phase change of $2\pi$. Therefore,

$$2\pi\Delta f\,\tau = 2\pi,$$

which gives

$$\tau = \frac{1}{\Delta f}$$

and

$$\boxed{v_p = 2\ell\Delta f}.$$

The velocity factor of the cable is

$$\boxed{\mathrm{VF} = \frac{v_p}{c}},$$

where $c \approx 3 \times 10^8\ \mathrm{m/s}$ is the speed of light in vacuum.

Here, $\Delta f$ is the frequency difference between two consecutive maxima or two consecutive minima.

### 1.3.3 Reflection coefficient and characteristic impedance

The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0},$$

where $Z_L$ is the load impedance and $Z_0$ is the characteristic impedance of the cable.

For important special cases:

- Open circuit: $Z_L \to \infty$ and $\Gamma = +1$.
- Short circuit: $Z_L = 0$ and $\Gamma = -1$.
- Matched load: $Z_L = Z_0$ and $\Gamma = 0$.

When the incident and reflected waves are in phase, they interfere constructively and the measured amplitude is maximum: $V_{\max} = |V_{\text{incident}}| + |V_{\text{reflected}}|$.

When the incident and reflected waves are out of phase by $180^\circ$, they interfere destructively and the measured amplitude is minimum. Assuming the incident-wave amplitude is greater than or equal to the reflected-wave amplitude,

$$V_{\min} = |V_{\text{incident}}| - |V_{\text{reflected}}|. \text{ Adding}$$

and subtracting these two equations gives

$$|V_{\text{incident}}| = \frac{V_{max} + V_{min}}{2}$$

And

$$|V_{\text{reflected}}| = \frac{V_{\max} - V_{\min}}{2}.$$

Therefore, the magnitude of the reflection coefficient observed at the Tee junction can be estimated as

$$|\Gamma| = \frac{|V_{reflected}|}{|V_{\text{incident}}|} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

In general, the characteristic impedance of a transmission line can be estimated by connecting different known terminations, such as $25\ \Omega$, $50\ \Omega$, $100\ \Omega$, an open circuit, and a short circuit, and measuring the corresponding reflected-signal amplitudes.

In this experiment, only a $50\ \Omega$ termination is used. The reflected signal obtained with this termination is compared with that obtained when the cable is left open. A significant reduction in the reflected-signal amplitude verifies that the cable is approximately matched to the $50\ \Omega$ termination and indicates that its characteristic impedance is close to $50\ \Omega$.

### 1.3.4 Inductance and capacitance per unit length

Let $L'$ denote the inductance per unit length in $\mathrm{H/m}$ and $C'$ denote the capacitance per unit length in $\mathrm{F/m}$. For a low-loss transmission line,

$$v_p = \frac{1}{\sqrt{L'C'}}$$

and

$$Z_0 = \sqrt{\frac{L'}{C'}}.$$

Therefore,

$$\boxed{L' = \frac{Z_0}{v_p}}$$

and

$$\boxed{C' = \frac{1}{Z_0 v_p}}.$$

### 1.3.5 Cable attenuation

Cable attenuation is the reduction in signal level as the wave propagates through the cable. In this experiment, power is measured in dBm using a spectrum analyser. The insertion loss of the cable is

$$IL = P_{\text{ref}} - P_{\text{cable}},$$

where $P_{\text{ref}}$ is the power measured using a short reference connection and $P_{\text{cable}}$ is the power measured after inserting the cable under test. The attenuation per unit length is then

$$\boxed{\alpha = \frac{IL}{\ell} \quad \mathrm{dB/m}}.$$

## 1.4 Components Required

- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector

- Coaxial cable under test
- Short reference coaxial cable
- $50\ \Omega$ termination
- Open-circuit termination, obtained by leaving the cable end unconnected
- Suitable RF adapters and connectors

## 1.6 Experimental Setup

Connect the signal generator, cable under test, and oscilloscope through the Tee connector as shown in . The far end of the cable is terminated according to the measurement being performed. For a fixed frequency, the oscilloscope normally displays a sinusoidal waveform. A mismatch changes the measured amplitude and phase at the Tee junction. When frequency is swept, the amplitude varies periodically because the round-trip phase of the reflected signal changes. A matched termination minimizes this amplitude variation.

## 1.7Experimental Procedures , Results and Observations.

### 1.7.1 Part A: Phase velocity and velocity factor

1. Measure and record the physical cable length $\ell$.
2. Connect the signal generator, Tee connector, cable, and oscilloscope as shown in Fig. 3.
3. Leave the far end of the cable open.
4. Set the oscilloscope input impedance to $1\ \mathrm{M}\Omega$.
5. Apply a sinusoidal signal of constant amplitude.
6. Sweep the frequency slowly over a range containing at least two maxima and two minima. Record the signal amplitude at suitable frequency intervals.
7. Plot the measured signal amplitude versus frequency and identify the frequencies $f_1$ and $f_2$ of two consecutive maxima or two consecutive minima.
8. Calculate the frequency spacing:

$$\Delta f = |f_2 - f_1|$$

Convert $\Delta f$ to hertz before using it in the phase-velocity calculation.

9. Calculate the phase velocity and velocity factor:

$$v_p = 2\ell\Delta f, \qquad \text{VF} = \frac{v_p}{c}.$$

Results and Observations :

$f_1 = 57$ mHz , $f_2 = 173$ mHz , l = 92.5cm

$$\Delta f = |f_2 - f_1| = |173\text{-}57| = 116 \text{ mHz}$$

$$v_p = 2\ell\Delta f = 2(92.5)(116) = 2.14 \text{ x } 10^8 \text{ m/s}$$

$$\text{VF} = \frac{v_p}{c} = 2.14 \text{ x } 10^8 / 3\text{x } 10^8 = 0.7133$$

### 1.7.2 Part B: Reflection coefficient and approximate characteristic impedance

1. Retain the Tee-junction setup and use the same frequency range employed for the opencircuit measurement in Part A.
2. From the open-circuit data, determine $V_{\text{max,open}}$ and $V_{\text{min,open}}$ over one complete ripple period.
3. Calculate the magnitude of the reflection coefficient for the open-ended cable:

$$|\Gamma_{\text{open}}| = \frac{V_{\text{max,open}} - V_{\text{min,open}}}{V_{\text{max,open}} + V_{\text{min,open}}}.$$

4. Connect the $50\ \Omega$ termination to the far end of the cable.
5. Sweep over the same frequency range without changing the signal-generator amplitude.
6. Record $V_{\text{max,50}}$ and $V_{\text{min,50}}$.
7. Calculate

$$|\Gamma_{50}| = \frac{V_{\text{max,50}} - V_{\text{min,50}}}{V_{\text{max,50}} + V_{\text{min,50}}}.$$

8. Compare $|\Gamma_{open}|$ and $|\Gamma_{50}|$. If the reflection is substantially reduced with the $50\ \Omega$ termination, conclude that the cable is approximately matched to $50\ \Omega$ and that its characteristic impedance is close to $50\ \Omega$.

$V_{max,open}$ = 1.72 V , $V_{min,open}$ = 125mV

$V_{max,50}$ = 928mV , $V_{min,50}$ = 900mV

$|\Gamma_{open}| = 1595/1845 = 0.8644$

$|\Gamma_{50}|$ = 28/1828 = 0.015

| Termination | $V_{max}$ (V) | $V_{min}$ (V) | $\|\Gamma\|$ |
|---|---|---|---|
| Open circuit | 1.72 | 0.125 | 0.8644 |
| $50\ \Omega$ load | 0.928 | 0.9 | 0.015 |

### 1.7.3 Part C: Inductance and capacitance per unit length

Using the measured $v_p$ and the approximate value $Z_0 \approx 50\ \Omega$ established from the matching test, calculate

$$L' = \frac{Z_0}{v_p}$$

and

$$C' = \frac{1}{Z_0 v_p}$$

Z0 = 50 x (1-0.015)/(1+0.015) = 48.52 Ω

L‘ = 48.52/2.146 x $10^8$ = 2.26 x $10^{-7}$/ 92.5 = $244 nH/m$

C’ = 1/48.5 x 2.146 x $10^8$ = 96.1 / 92.5 = $103.89$ pF/m

### 1.7.4 Part D: Cable attenuation

1. Connect the signal generator to the spectrum analyser using a short reference cable and the required adapters.
2. Set a suitable test frequency and input power. Keep both fixed throughout the measurement.
3. Record the reference power $P_{\text{ref}}$ in dBm.
4. Replace the short reference connection with the cable under test while keeping the remaining setup unchanged.
5. Record the measured power $P_{\text{cable}}$ in dBm.
6. Calculate the cable insertion loss:

$$IL = P_{\text{ref}} - P_{\text{cable}}.$$

7. Calculate the attenuation per unit length:

$$\alpha = \frac{IL}{\ell} \text{ dB/m}.$$

8. Repeat at additional frequencies if the frequency dependence of attenuation is to be studied.

| Frequency (MHz) | $P_{\text{ref}}$ (dBm) | $P_{\text{cable}}$ (dBm) | $\alpha$ (dB/m) |
|---|---|---|---|
| 100 | -0.57 | -0.60 | 0.032 |
| 150 | -0.79 | -0.72 | 0.032 |
| 200 | -0.91 | -0.90 | 0.01 |

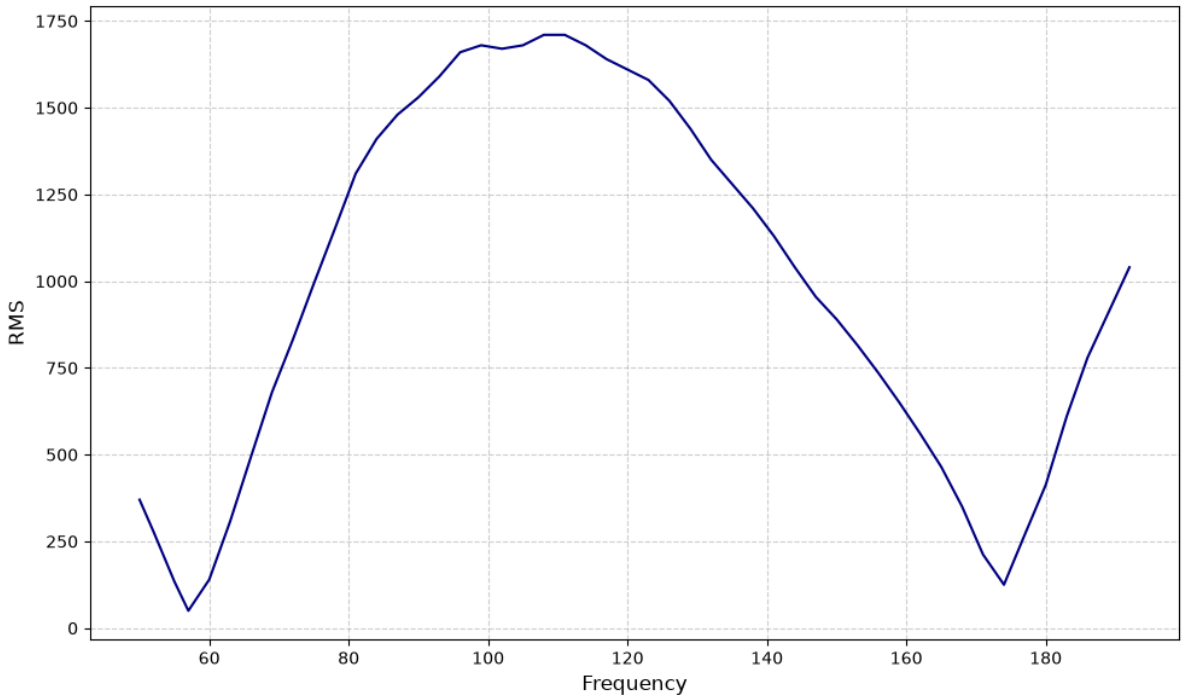
## 1.8 Results and Observations

Summarize the measured and calculated quantities in the following table.

| Quantity | Symbol | Measured/Calculated Value |
|---|---|---|
| impedance (Ω) | $\ell$ | 92.5 |
| | $\Delta f$ | |
| | $v_p$ | 116 |
| | $v_p/c$ | 0.7133 |
| | $Z_0$ | 48.52 |

| | | |
|---|---|---|
| | | |
| Inductance per unit length (H/m) | $L'$ | 22.6 |
| Capacitance per unit length (F/m) | $C'$ | 96.1 |
| Attenuation (dB/m) | $\alpha$ | 0.032 |

- the plot of measured standing-wave signal amplitude versus frequency for the open-ended cable;



- the calculation of $\Delta f$, $v_p$, and velocity factor.

  Calculations ae mentioned above in procedure , results and observations.

- the open-circuit and $50\ \Omega$ amplitude-ripple measurements, including the calculation and comparison of $|\Gamma_{\text{open}}|$ and $|\Gamma_{50}|$;

  Calculations ae mentioned above in procedure , results and observations.

- the calculation of $L'$ and $C'$;

  Calculations ae mentioned above in procedure , results and observations.

- the reference and cable power measurements used for attenuation;

  Calculations ae mentioned above in procedure , results and observations.

## 1.9 Conclusions

State the following results and briefly discuss their physical significance:

- plot showing the variation of standing-wave signal amplitude with frequency due to interference.

  The measured standing-wave signal amplitude was plotted as a function of frequency. The amplitude showed periodic maxima and minima due to interference between the incident and reflected waves in the cable. The positions of the maxima and minima depend on the phase difference between these waves, which changes with frequency. Thus, the observed periodic variation confirms the standing-wave behaviour of the transmission line.

- measured phase velocity and velocity factor .

  The phase velocity was determined from the measured spacing between successive interference maxima/minima.

  The velocity factor indicates how fast electromagnetic waves propagate through the cable compared with propagation in free space. A value less than unity is expected because the cable contains a dielectric medium.

- verification that the cable is approximately matched to a $50\ \Omega$ termination, indicating that its characteristic impedance is close to $50\ \Omega$.

  The cable was terminated with a (50 Ω) load and the standing-wave behaviour was examined. The relatively small reflection under this condition indicates that the load is approximately matched to the characteristic impedance of the cable.

  • calculated inductance and capacitance per unit length;
  L' = 22.6$\mu$H , C' = 96.1pF . Capacitance and inductance per unit length define how a transmission line stores electrical and magnetic energy along its physical path, dictating how signals travel through cables and wires.

  • measured attenuation per unit length;

  The attenuation was determined by measuring the reduction in signal amplitude/power as the signal propagated through the cable. The measured attenuation represents the loss of signal strength due to resistive losses in the conductors and dielectric losses in the insulating material. Lower attenuation indicates more efficient transmission of the signal.

- possible sources of measurement uncertainty :

  • Uncertainty in measuring the cable length.
  • Frequency resolution and accuracy of the signal generator.
  • Uncertainty in identifying the exact positions of maxima and minima.
  • Measurement uncertainty of the oscilloscope/voltmeter.
  • Imperfect (50,\Omega) termination and connector mismatch.
  • Losses and reflections at connectors and joints.
  • Cable bending or changes in the cable geometry.
  • Variation in signal amplitude with frequency from the generator or measurement system.
  • Instrument noise and environmental electromagnetic interference.
  • Uncertainty in voltage/amplitude measurements used to calculate attenuation.